

Clustering Analysis of 2020 Daily Stock Returns

Data and Similarity Measure

The data set records the daily return rate of 22 listed stocks over the course of 2020 ($n = 251$ trading days, no missing values). The stocks span banking (JPM, Citi, WF), energy (Shell, Exxon, Chevron), industrials (Dow, Eastman, Lyondell, DuPont, Honeywell), pharmaceuticals (Pfizer, J&J, Merck, Eli Lilly), technology (Apple, Nvidia, Microsoft, Google, Adobe), and two idiosyncratic stocks (Moderna, Kodak).

The sample Pearson correlation coefficient $r(i, j)$ is used to calculate similarity in each pair of stocks (i, j) . The corresponding dissimilarity $d(i, j) = 1 - r(i, j)$ is used where $d(i, j) \in [0, 2]$ as a valid dissimilarity matrix used to create average, complete and single linkage clusters as well as PAM clusters. For Ward's method, which utilizes a Euclidean-compatible distance metric, the chord distance $d_{\text{chord}}(i, j) = \sqrt{2 * (1 - r(i, j))}$ is utilized to replace the original dissimilarity matrix because it represents the Euclidean distance between the standardized return series projected onto the surface of a unit sphere.

Most stock return series have positive correlation coefficients. The median pairwise correlation coefficient is 0.52 while the mean pairwise correlation coefficient is 0.5 and reflect widespread co-movements during the COVID-19 pandemic year. Two stocks are exceptions to these positive correlations. Kodak has both the highest mean daily standard deviation ($\sigma = 0.252$) and the highest single day return (0.318) due to an extreme price increase on a single trading day following a July 2020 U.S. government loan announcement; Kodak has one of the lowest mean correlations with all other stocks (0.078). Moderna has the second-highest mean daily standard deviation ($\sigma = 0.0640$) and the lowest mean correlation with all other stocks (0.002) since most of Moderna's returns can be attributed to announcements related to vaccine trials unrelated to general market movements. The five strongest pairwise correlations are found to be JPMorgan/Citigroup (0.917), Dow Chemical/LyondellBasell (0.910), JPMorgan/Wells Fargo (0.893), Microsoft/Adobe (0.874), and Exxon/Mobil Chevron (0.866) - all intra-industry pairs.

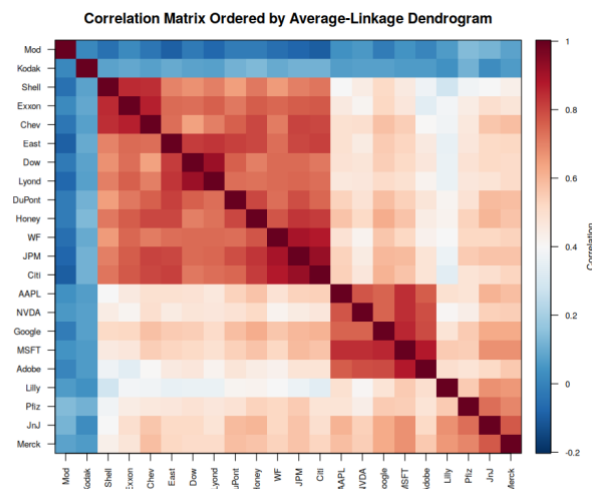


Figure 1. Correlation matrix with rows and columns reordered by the average-linkage dendrogram leaf order. The block structure matches the five-cluster solution.

Clustering Methods

The average, complete and single linkage methods are all applied to $d(i, j) = 1 - r(i, j)$; the other hierarchy method is Ward's method using the chord distance as the measure for the clustering. In addition to these four hierarchical measures, PAM (Partitioning Around Medoids) is also run to serve as a non-hierarchical validation of the hierarchical results using $d(i, j) = 1 - r(i, j)$ as the similarity function.

Cophenetic correlations (Table 1) show that average linkage best preserves the original distances ($r = 0.97$).

Ward's lower value (0.66) reflects its different objective — minimizing within-cluster variance — rather than poor cluster quality.

Method	Cophenetic Correlation
Average (UPGMA)	0.973
Single	0.959
Complete	0.939
Ward (chord)	0.656

Table 1. Cophenetic correlation for each linkage method (all measured against $d = 1 - r$ for comparability).

Figure 2 shows the dendrogram for each method along with rectangles representing the grouping as per $k = 5$. Despite differences in branch heights and tree shapes, all four methods produce the same five subgroups at $k = 5$, suggesting these represent real patterns in the data.

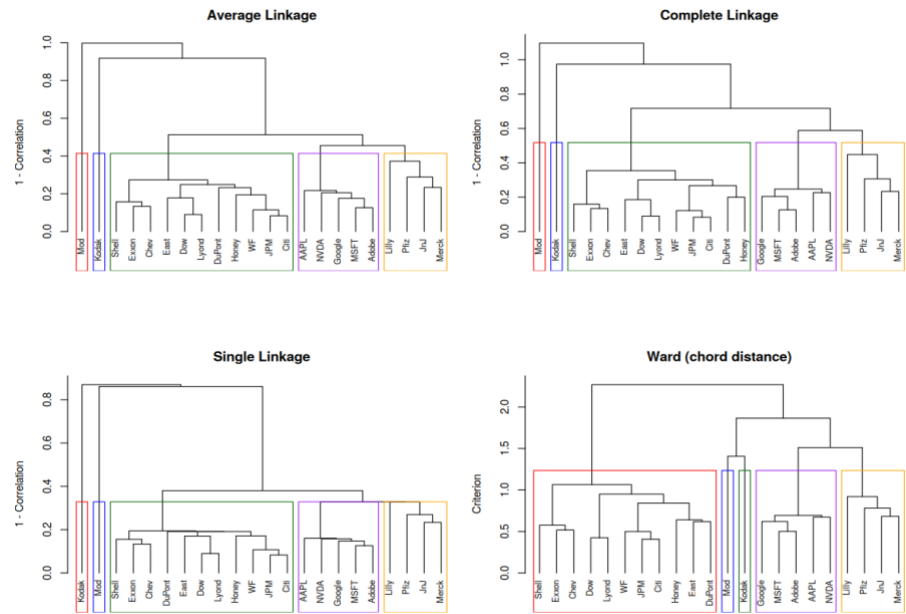


Figure 2. Dendrograms for all four linkage methods with $k = 5$ clusters highlighted. Despite differences in branch heights, the five-cluster partition is consistent across all methods.

Choosing the Number of Clusters

Mean silhouette widths for $2 \leq k \leq 7$ are shown in Figure 3. Average linkage peaks at $k = 2$ (0.51) and Ward at $k = 2$ (0.33); both reach a secondary maximum at $k = 5$ (0.44). PAM also yields 0.44 at $k = 5$. Although $k = 2$ scores higher, it merely separates the two outliers from the remaining 20 stocks and reveals no sector structure. We select $k = 5$ as the primary solution for its stability across all methods and interpretable alignment with economic sectors.

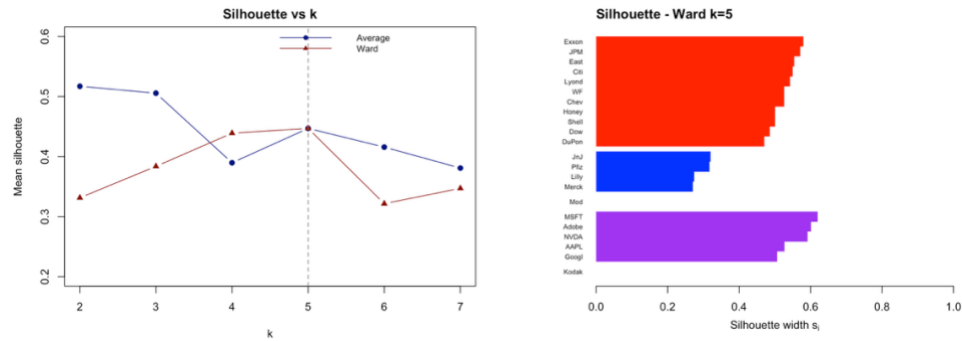


Figure 3. Left: mean silhouette width vs. k for average linkage and Ward. Both peak at $k = 2$ but reach a secondary maximum at $k = 5$ (dashed line). Right: silhouette plot for Ward $k = 5$; most stocks show positive widths, confirming appropriate assignment.

Final Clustering Solution ($k = 5$)

Table 2 summarizes the five-cluster solution. PAM ($k = 5$) produces the same partition, with medoids JPM, J&J, Moderna, Microsoft, and Kodak.

Cluster	Members (22 stocks)	Interpretation
1	JPM, Citi, WF, Shell, Exxon, Chev, Dow, East, Lyond, DuPont, Honey	Cyclical/industrial: banking, energy, chemicals (within-cluster mean $r \approx 0.76$)
2	Pfizer, JnJ, Merck, Lilly	Large pharmaceuticals: moderate inter-correlations, insulated from energy/banking sell-off.
3	Moderna	Idiosyncratic biotech: vaccine-trial driven, mean $r \approx 0.002$.
4	AAPL, NVDA, MSFT, Google, Adobe	Technology: tightly co-moving growth stocks (within-cluster mean $r \approx 0.80$).
5	Kodak	Distressed outlier: extreme July 2020 price event, mean $r \approx 0.078$.

Table 2. Five-cluster solution. Membership is identical for all four hierarchical methods and PAM.

Figure 4 confirms the five-cluster structure in two dimensions: the first coordinate separates the outliers (Moderna, Kodak) from the main body; the second separates technology from cyclical stocks, with pharmaceuticals in between.

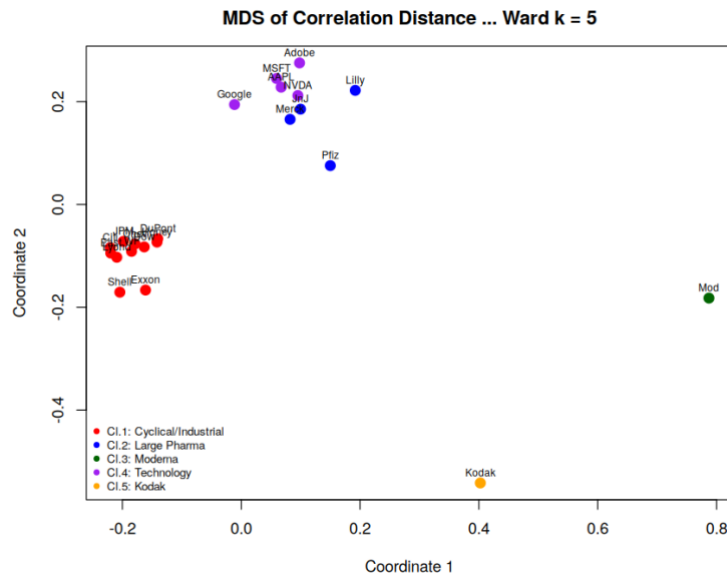


Figure 4. Classical MDS of the correlation-distance matrix, colored by Ward $k = 5$ cluster membership.

Conclusion

In addition to recovering five clusters using all five methods (as shown above) we find a consistent five cluster solution for correlation-based cluster analysis of 22 stock prices. These five clusters correspond to: Industrial, Large Pharmaceuticals, Technology, Moderna and Kodak. Although $k = 2$ yields a higher silhouette score, we choose the $k=5$ solution as it shows meaningful sectors versus simply separating out the two outlier companies.

Appendix: R Code

Loading data

```
library(cluster)

d <- read.table("stocknew.d", header = TRUE)
stock_labels <- c("JPM", "Citi", "WF", "Shell", "Exxon", "Chev", "Dow", "East",
                  "Lyond", "DuPont", "Honey", "Pfizer", "JNJ", "Mod", "Merck",
                  "Lilly", "AAPL", "NVDA", "MSFT", "Google", "Kodak", "Adobe")
```

Correlation matrix & dissimilarity

```
cor_mat <- cor(d)
dist_cor <- as.dist(1 - cor_mat)
chord_dist <- as.dist(sqrt(2 * (1 - cor_mat))) #used for Ward
```

Standard deviations of daily returns

```
sds <- apply(d, 2, sd)
sort(round(sds, 4), decreasing = TRUE)
```

Top correlated pairs

```
upper_pairs <- which(upper.tri(cor_mat), arr.ind = TRUE)
top <- data.frame(s1 = rownames(cor_mat)[upper_pairs[, 1]],
                 s2 = colnames(cor_mat)[upper_pairs[, 2]],
                 corr = cor_mat[upper_pairs])

head(top[order(-top$corr), ], 8)
```

Avg correlation of each stock with others

```
avg_cor <- (rowSums(cor_mat) - 1)/(ncol(cor_mat) - 1)
sort(round(avg_cor, 3)) # Moderna and Kodak near zero
```

Hierarchical clustering

```
hc_avg <- hclust(dist_cor, method = "average")
hc_comp <- hclust(dist_cor, method = "complete")
hc_sing <- hclust(dist_cor, method = "single")
hc_ward <- hclust(chord_dist, method = "ward.D2")
```

Cophenetic correlations

```
sapply(list(avg = hc_avg, comp = hc_comp,
            sing = hc_sing, ward = hc_ward),
       function(h) round(cor(dist_cor, cophenetic(h)), 4))
```

Silhouette analysis (k = 2..7)

```

sils_avg <- sapply(2:7, function(k) mean(silhouette(cutree(hc_avg, k), dist_cor)[,
3]))
sils_ward <- sapply(2:7, function(k) mean(silhouette(cutree(hc_ward, k), dist_cor)[
, 3]))
names(sils_avg) <- names(sils_ward) <- 2:7

round(rbind(average = sils_avg, ward = sils_ward), 3)

```

PAM

```

pm <- pam(dist_cor, k = 5, diss = TRUE)
stock_labels[pm$id.med]

mean(pm$silinfo$widths[,3])

```

Final clusters from Ward

```

ccol <- c("red", "blue", "darkgreen", "purple", "orange")
cl <- cutree(hc_ward, k = 5)
split(stock_labels, cl)

```

Fig 1 - heatmap

```

ord <- hc_avg$order
par(mar = c(6,6,3,2))
image(1:22, 1:22, cor_mat[ord, rev(ord)],
      col = hcl.colors(50, "RdBu", rev=TRUE), zlim = c(-0.2, 1),
      axes = FALSE, xlab = "", ylab = "",
      main = "Correlation matrix (avg linkage order)")
axis(1, 1:22, stock_labels[ord], las=2, cex.axis=0.6)
axis(2, 1:22, rev(stock_labels[ord]), las=2, cex.axis=0.6)
box()

```

Fig 2 - dendrograms

```

par(mfrow=c(2,2), mar=c(4,4,2,1))
for(hc in list(hc_avg, hc_comp, hc_sing, hc_ward)) {
  nm <- c(average="Average", complete="Complete",
          single="Single", ward.D2="Ward (chord)")[hc$method]
  yl <- if(hc$method == "ward.D2") "Criterion" else "1 - r"
  plot(hc, labels=stock_labels, main=nm, xlab="", sub="", ylab=yl, cex=0.7, hang=-1
  )
  rect.hclust(hc, k=5, border=ccol)}

```

Fig 3 - silhouette

```

par(mfrow=c(1,2), mar=c(4,4,2,1))
plot(2:7, sils_avg, type="b", pch=19, col="navy", ylim=c(0.2, 0.6),
     xlab="k", ylab="Mean silhouette", main="Silhouette vs k")
lines(2:7, sils_ward, type="b", pch=17, col="darkred")
abline(v=5, lty=2, col="gray50")
legend("topright", c("Average", "Ward"), col=c("navy", "darkred"),

```

```

        pch=c(19,17), lty=1, cex=0.8, bty="n")
par(mar=c(4,8,2,1))
sil_obj <- silhouette(cl, dist_cor)
rownames(sil_obj) <- stock_labels
plot(sil_obj, main="Silhouette - Ward k=5",
      col=ccol, border=NA, cex.names=0.6, do.clus.stat=FALSE, do.n.k=FALSE)

```

Fig 4 - MDS

```

mds <- cmdscale(dist_cor, k=2)
par(mfrow=c(1,1), mar=c(4,4,2,1))
plot(mds[,1], mds[,2], col=ccol[cl], pch=19, cex=1.4,
      main="MDS of correlation distance", xlab="Dim 1", ylab="Dim 2")
text(mds[,1], mds[,2], stock_labels, cex=0.65, pos=3)
legend("bottomleft",
      c("Cyclical", "Pharma", "Moderna", "Tech", "Kodak"),
      col=ccol, pch=19, cex=0.7, bty="n")

```